

Data Collection and Preprocessing Phase

Date	Aug 18, 2026
Team ID	Individual
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 345,407 records and 10 attributes related to agricultural production in India from 1997 to 2021 . The main attributes include State, District, Crop, Year, Season, Area, Area Units, Production, Production Units, and Yield-related data. The dataset is used to analyze agricultural performance across different states, crops, seasons, and years.
Data Cleaning	The dataset was checked for missing values and duplicate records. No missing values and no duplicate records were found. Column names were reviewed for consistency, including standardizing the season field name to Season. A worksheet naming typo, Distrinution, was also identified and corrected/recommended for correction to Distribution.
Data Transformation	The data was filtered and aggregated in Tableau to support different analyses based on Crop, State, Year, and Season . Top-5 comparisons were used for selected analyses, including leading states by agricultural production and major crops or states based on cultivated area. Measures such as Area, Production, and Yield were aggregated to create comparative and trend-based visualizations.
Data Type Conversion	Numeric agricultural measures such as Area, Production, and Yield were used as numerical fields. The Year field was reviewed for chronological analysis and used in year-wise

	visualizations. No explicit calculated data-type conversion formula was identified in the final Tableau workbook.
Column Splitting and Merging	No column splitting or merging operations were required or explicitly identified in the dataset/workbook. The available columns were already structured separately for dimensions such as State, District, Crop, Season, and Year.
Data Modeling	The project uses a single primary agricultural dataset/data source for analysis. Dimensions such as Area, Production, and Yield are used with dimensions including State, District, Year, and Season to create Tableau worksheets, dashboard visualizations, and the final Tableau Story. No multi-table relationship or join structure was identified.
Save Processed Data	After preprocessing and organizing the data for analysis, the dataset was used to create the Tableau workbook and embedded as a Tableau Hyper extract . The processed data was then used across the worksheets, main dashboard, and Tableau Story for further analysis and visualization.